Determining Fair CDS Premiums from Default Probabilities

The Assignment

Your analysts obtain the following information:

- The set of default probabilities for XYZ plc for each six-month period for the next five years
- The set of six-month forward rates

(You may assume that each six-month period is exactly 0.5 year long.)

Your analysts also estimate that the recovery rate is 40%.

Period	Forward rates	Default probability PD_j
0.5	2.25%	6.00%
1.0	2.50%	5.50%
1.5	2.88%	5.00%
2.0	3.00%	4.50%
2.5	3.25%	4.00%
3.0	3.50%	3.50%
3.5	4.00%	3.00%
4.0	4.38%	2.50%
4.5	4.75%	2.00%
5.0	5.00%	2.00%

REQUIRED

Using the spreadsheet template supplied, calculate:

1. The cumulative survival probabilities at the end of each six-month period.
2. The cumulative survival-probability-weighted present values of the receipts of a $1 p.a. CDS premium (equivalent to a 100% p.a. CDS premium on a notional of $1).
3. The cumulative default-probability-weighted present values of the default payments.
4. The set of fair CDS premiums for CDS for each maturity from 6m to 5y.

USEFUL FORMULA

The following formula gives you the value of the swap premium:

$$s_N = \frac{(1-R)\sum_{j=1}^{N} v_j PD_j}{\sum_{j=1}^{N} v_j PS_j T_{j-1,j}}$$

Where:

v_j	is the discount factor for period j
PD_j	is the probability of default in period j
PS_j	is the probability of survival through period j
$T_{j-1,j}$	is the length of time of period j
R	is the recovery rate

© 2004-22 ACF Consultants Ltd
All rights reserved